Nathan Houck

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EXECUTIVE SUMMARY

Aerospace engineering student with a strong foundation in astronautics and practical experience in thermodynamics and heat transfer. Seeking a Summer 2026 aerospace engineering internship to apply technical expertise and and classroom knowledge in spacecraft systems, thermal management, or propulsion design.

EDUCATION

Embry-Riddle Aeronautical University — Daytona Beach, FL
Bachelor of Science, Aerospace Engineering — May 2027
Area of Concentration: Astronautics — GPA: 3.75/4.0
Dean's List, Fall 2023, Spring 2024, Spring 2025, Fall 2025
Honor Roll, Fall 2024
Exceptional Eagles Award – ERAU Student Involvement Awards, Spring 2025

PROJECT EXPERIENCE

Autonomous Maritime Robotics Association – Robosub: Aug. 2024 – Aug. 2025

Heat Transfer Team Lead

- Researched a solution for electronics cooling within the submarine to reduce internal temperature by over 60°F
- Investigated heat pipes and thermoelectric cooling.

Mechanical Engineering Team

- Worked with a team of 4 to design, prototype, and manufacture an improved thruster mount.
- Designed and modeled a torpedo launching system that launches the torpedoes when the proper target is identified.
- Prototyped a bin that will release weighted bags over a designated drop zone, utilizing torsion springs and servo motors.

Experimental Jet Engine Propulsion Club (XJEP): Aug. 2025 – Feb. 2026

Aerospace Engineer

- Designing a novel cooling solution for a modular afterburner attached to a KingTech turbine engine.
- Working on a paper that will be published in AIAA's SciTech convention studying afterburner efficiency.

Experimental Aerodynamics Data Processing: Jan. 2026 – Apr. 2026

- Developed MATLAB scripts to clean, filter, and normalize raw sensor data from wind tunnel sensor data.
- Applied aerodynamic principles to interpret experimental results and draw meaningful conclusions.
- Formulated a hypothesis and designed an experiment to evaluate the effect of the boundary layer "tripping" on a NACA 0012 airfoil.

RESEARCH

Undergraduate Researcher – Thermal Science Laboratory: Aug. 2025 – Dec. 2025

- Collaborated with the Insituto Technológico de Aeronáutica to create a flow visualization model to study the behavior of phase change materials.
- Created a test section using Autodesk Inventor to contain the PCM and allow it to be oriented at any angle.
  - The test section simulated a constant heat flux boundary condition and a constant

SKILLS

Data Analysis: Python, Claude, ChatGPT, Gemini, and emerging technologies
Engineering Software: CATIA v.5, Autodesk Fusion, MATLAB, Java, C/C++ (Basic)
Technical: Drafting, 3-D printing
Languages: English (fluent), German (basic)
Hobbies and Interests: Scuba Diving (Open Water and Nitrox), Aviation, Guitar, Alto Saxophone, Fossils

ACTIVITIES

Conservation Officer, Studying Habitats and Reef Kingdoms, Spring 2026 – Present

- Spearheading new conservation program looking to contribute to Florida's coral conservation efforts
- Developing local programs around mangrove planting and artificial reef monitoring
- Active diver (45+ dives) with international experience

Member, Experimental Jet Engine Propulsion Club, Fall 2025 – Present
Flight Student, Summer 2024 – 2026

- Risk management and avoidance
- Effective communication and situational awareness
- Problem solving and remaining calm under pressure
- 28.1 Flight Hours (not current)

Volunteer, Event Supervisor, Science Olympiad Regionals, Fall 2019 – Present